Mapping Precipitable Water Vapor Time Series From Sentinel-1 Interferometric SAR

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***Abstract*— In this article, a methodology to retrieve the precipitable water vapor (PWV) from a differential interferometric time series is presented. We used external data provided by atmospheric weather models (e.g., ERA-Interim reanalysis) to constrain the initial state and by Global Navigation Satellite System (GNSS) to phase ambiguities elimination introduced by phase unwrapping algorithm. An iterative least-square is then used to solve the optimization problem. We applied the presented methodology to two time series of differential PWV maps estimated from synthetic aperture radar (SAR) images acquired by the Sentinel-1A, over the southwest part of the Appalachian Mountains (USA). The results were validated using an independent GNSS data set and also compared with atmospheric weather prediction data. The GNSS PWV observations show a strong correlation with the estimated PWV maps with a root-mean-square error less than 1 mm. These results are very encouraging, particularly for the meteorology community, providing crucial information to assimilate into numerical weather models and potentially improve the forecasts.**

***Index Terms*— Global navigation satellite system (GNSS), interferometric synthetic aperture radar (InSAR), precipitable water vapor (PWV), Sentinel-1, synthetic aperture radar (SAR), time series.**

I. Introduction

THE interferometric synthetic aperture radar (InSAR) technique have been used to provide maps of differential precipitable water vapor (PWV) (ΔPWV) with high precision and spatial resolution [1]–[8]. The Sentinel-1A/B satellites acquire synthetic aperture radar (SAR) images with a large spatial coverage (length of hundreds of kilometers by 250 km) mean revisiting time of 6 days which can be reduced to 1 day in specific regional areas and combining different orbits [9]. This allows imaging of the atmospheric water vapor variations over short spatial scales (<10 km) boosting the concept of "InSAR Meteorology" in the geodetic and meteorological community. Recently, some authors have assimilated InSAR maps PWV into high-resolution Numerical Weather Prediction (NWP) models, improving the forecast of light to extreme rainfall events [5], [9], [10]. From the atmospheric point of view, the main limitation to the use of InSAR data is the estimation of PWV spatial distribution at the acquisition time of the first SAR image, also known as the master image in the case of a single interferometric couple. Several studies have addressed this problem. Alshawaf *et al*. [11] applied the least squares (LSQ) method to retrieve the nondifferential PWV corresponding to each SAR acquisition time. The system is rank deficient if there are $N-1$ observations (interferograms) and N unknown (images). To overcome this problem, they added a conditioning constraint to the system, assuming that the mean of all observations is close to zero. This approach has two disadvantages. First, the observations (phase) are considered to be a zero-mean Gaussian process. Second, the constraint implies a large interferometric time series, which is not always needed in real applications. Pinchelli *et al*. [5] used external sources, as the Medium Resolution Imaging Spectrometer (MERIS) which was on board of the inactive ENVISAT, supplying simultaneously optical images to the SAR acquisitions in order to estimate the expected nondifferential value and assuming that the SAR and the external source are observing the same atmosphere. However, external observations close to the SAR acquisition times and with a spatial resolution in the order of tens of kilometers are very unlikely. Moreover, other satellite multispectral instruments capable of observing water vapor content as the Moderate-Resolution Imaging Spectroradiometer (MODIS) or Atmospheric Infrared Sounder (AIRS), are affected by cloud coverage, lack of sunlight, or discrepancies in spatial resolution and/or time overlapping. Mateus *et al*. [9], [10] employed an NWP model (e.g., the Weather and Forecast Model) to provide a field of PWV with a high spatial resolution ($\leq$1 km) at SAR master acquisition time, obtaining the absolute value at SAR slave time. An independent Global Navigation Satellite System (GNSS) data set was used to evaluate the absolute signal, obtaining a root-mean-square error (RMSE) lower than 2 mm. However, NWP models at these spatial resolutions run only for small areas, requiring large computational resources, and for this reason, this method is not widely used. To overcome these limitations, we propose a sequential constrained LSQ linear model to derive maps of PWV using a time series (sequence) of SAR interferograms. The LSQ model requires an initial map of PWV that can be provided by a numeric weather model and a minimum of one observation of the water vapor in the area covered by the interferograms, which could be provided by a GNSS station.The proposed method is applied to two time series of consecutive SAR images acquired by Sentinel-1A over the southwest part of the Appalachian Mountains (USA).

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The southwest part of the Appalachian Mountains is characterized by a humid continental climate, with frequent rainfall events strongly modulated by the complex topography, resulting in high spatial and temporal variability of the water vapor variability [12]. A significant fraction of its rainfall (30%–50%) is associated with light rain from the fog and low-level clouds [13]. The mountains system acts as a topographic barrier to tropical air mass (warm and moist) that coming from the Gulf of Mexico, and to continental and polar air mass (cold and dry), generating a complex spatial–temporal atmospheric moisture dynamic around and over the mountains, therefore being a suitable region to test this methodology. The results are assessed using an independent set of GNSS stations. We also evaluate the spatial variability of the InSAR maps of PWV and compare it to the ERA-Interim model. This allows studying the spatial variability of PWV maps provided by InSAR and statistically compare them to the same PWV property of the ERA-Interim model. This article is organized as follows. Section II is devoted to a short introduction to the physics of the atmosphere. Sentinel-1, GNSS, and ERA-Interim data used in this study are described in Section III. The method is explained in Section IV, and the results are present in Section V. Finally, in Section VI, we draw a few conclusions.

II. Physics of Microwave Phase Delay

The troposphere refractive index depends on the water vapor content, pressure, and temperature variations, influencing the velocity of the electromagnetic signals. In InSAR, this slowing down of the waves has an effect on the measured phase of the signal. This is not a problem for single-pass interferometry, when both SAR images are acquired simultaneously, but is a problem for repeat-pass due to the changes in the water vapor concentrations during the time elapsed between passes. The path delay of the electromagnetic signal due to propagation in the atmosphere can be written as

$$\delta R = \int (n-1) ds \tag{1}$$

where n is the tropospheric refractive index, ds is an infinitesimal part of the path, the integration is taken along the signal path in the troposphere, and δR is the path delay. The refractive index anomaly $(n-1)$ is usually scaled by

$$N = 10^6 \cdot (n-1) \tag{2}$$

where N is called refractivity. It has three components: dry (N_{dry}), wet (N_{wet}), and liquid (N_{liq}). The dry part contributes to about 90% of the total, is due to the dry atmosphere, and can be calculated with high accuracy ($\leq$ 1 mm). The wet and liquid components are related to tropospheric water vapor and liquid droplets in clouds, respectively. Both are not easy to model due to their spatial and temporal variability. Also, it is not possible to disentangle the effects of wet and the liquid temporal changes on the propagation delay without external measurements. Since the electromagnetic signal travels through the ionospheric layer, a fourth contribution has to be added to mitigate the ionospheric effect (N_{ion}). Thus, the total refractivity can be expressed as

$$N = N_{\text{dry}} + N_{\text{wet}} + N_{\text{liq}} + N_{\text{ion}} \tag{3}$$

and approximated by

$$N = k_1 \frac{P_d}{T} + \left(k_2 \frac{e}{T} + k_3 \frac{e}{T^2}\right) + 1.45W - 4.028 \times 10^7 \frac{n_e}{f^2} \tag{4}$$

where $k_1 = 77.689$ K hPa^{-1}, $k_2 = 71.2952$ K hPa^{-1}, and $k_3 = 3.75463 \times 10^5$ K^2 hPa^{-1} are called refractivity constants [14]; P_d, T, and e are, respectively, the pressure due to the dry air (in hPa), the temperature (in kelvins), and the water vapor pressure (in hPa); W is the liquid water content (in g m^3); n_e is the electron density per cubic meter; and f is the radar frequency [15]. The total delay is obtained by integrating (4) along the signal path. The dry part of refractivity can be calculated using surface measurements or numerical weather models data since it is not related to turbulent atmospheric parameters [16]. The ionospheric refractivity can be mitigated using maps of total electrons content (TEC) available globally with 6-h frequency. The accuracy of each PWV map has been assessed by comparing InSAR and GNSS measurements of PWV estimated using the fourteen permanent GNSS station [15]. A maximum RMSE of 2 mm has been found. This figure is changing along the year depending on vegetation decorrelation and can be as small as 1 mm. In this study, the mathematical model for both, the dry and ionospheric part, can be found in [8], [15], and [9]. The delay induced by the liquid refractivity is very small, less than 0.1-0.4 mm km^{-1} under usual conditions and about 0.7-3.1 mm km^{-1} in cumulus congestus and cumulonimbus [15], and it can be neglected. Therefore, the wet delay can be approximated by

$$\delta R_{\text{wet}} = 10^{-6} \cdot \left(k_2' \int \frac{e}{T} dz + k_3 \int \frac{e}{T^2} dz\right) \tag{5}$$

where $k_2' = k_2 - m k_1$ and m is M_w/M_d the ratio of the molar masses of water vapor and dry air [17]. We can relate the PWV to the wet delay

$$\text{PWV} = \Pi \cdot \delta R_{\text{wet}}. \tag{6}$$

Π depends on mean temperature profiles, varying with season, weather, and specific location, which induces the value of Π to vary by as much as 10% [18]. As a rule of thumb, $\Pi \approx 0.15$ [19].

III. Data

A. InSAR

A set of 24 single-look complex images with 5×20 m^2 spatial resolution, acquired by Sentinel-1A in interferometric wideswath (IW) mode over the southwest part of the Appalachian Mountains (USA), along an ascending orbit, was used. The footprint area of images is 170×255 km^2 (see Fig. 1). In this study, we used images acquired in two separated time intervals. The first one between October 2 and December 1, 2016 and the second between January 30 and April 12, 2017. All interferograms but one have a 12-day temporal baseline minimizing the effects of surface deformation and loss of temporal coherence. Precise orbits and an external high-resolution digital elevation model (30 m) were used to model and remove the orbital and topographic phase contributions, respectively. A 152×25 multilook-filter

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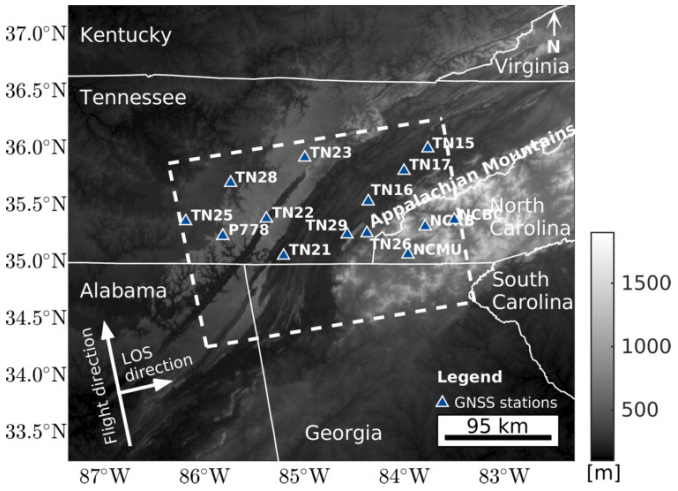


Fig. 1. Study area, southwest part of the Appalachian Mountains (USA). Dashed rectangle: Sentinel-1A footprint. Triangles: permanent GNSS network location. Gray background: elevation.

was applied to reduce phase noise so reducing the ground spatial resolution to 350×350 m^2. Pixels with coherence smaller than 0.2 were masked out to improve the phase unwrapping algorithm accuracy. A linear interpolation step was implemented at the end. The dry and ionospheric propagation delays (5) were modeled and removed resulting in the temporal change of the PWV

$$\Delta \mathrm{PWV} = \Pi \cdot \ \delta R_{\mathrm{wet}} \cdot \cos(\theta_{\mathrm{inc}}) \tag{7}$$

where λ is the radar wavelength, δR_{wet} is the temporal change of propagation delay due to water vapor, and θ_{inc} is the incidence angle [15].

B. GNSS

We used a network of 14 permanent GNSS stations belonging to the Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC) network, administrated by the University Corporation for Atmospheric Research (UCAR) [20], and to the Continuously Operating Reference Stations (CORS) network, administrated by the National Oceanic and Atmospheric Administration's (NOAA) National Geodetic Survey (NGS) [21]. Since the atmospheric water vapor is characterized by a high spatial and temporal variability, we decided to process the GNSS raw data, in order to obtain a solution as close as possible to the acquisition time of the SAR images (about 23:45 UTC).

The GAMIT/GLOBK package was used to estimate PWV with a precision of about 1 mm [22]. Fig. 1 shows the network of permanent GNSS stations processed. The GNSS estimates were used as a constraint in the proposed methodology.

C. Model Data

The ERA-Interim model [23] was used to derive the PWV maps needed for both estimating the PWV spatial distribution at the master time and, as an independent PWV map, to assess the results of the proposed methodology. Even if other numerical models can be used, we selected ERA-Interim as: 1) it is a reanalysis of the global atmosphere since 1979; 2) it updates of the archive take place on a monthly basis after a quality assurance and correction of technical problems with the production; and 3) data are available for any area in the world and can be freely downloaded with a time frequency of 6 h and a spatial spacing of 0.125°×0.125°, obtained from the 0.75°×0.75° native resolution after a bilinear interpolation [24]. An optimal interpolation (OI) algorithm was used to merge the GNSS PWV values and the ERA-Interim PWV maps improving its spatial detail [10].

IV. Methodology

Given a time series of N interferograms generated in order to minimize the temporal baseline, the PWV at time t can be computed as follows [see (7)]:

$$\mathrm{PWV}_{i,j}^{t+1} = \mathrm{PWV}_{i,j}^{t} + \Pi \cdot \delta R_{i,j} \cdot \cos(\theta_{i,j}) \tag{8}$$

where (i,j) are the pixel coordinates, $\mathrm{PWV}_{i,j}^{t+1}$ is the water vapor at date t+1, and $\mathrm{PWV}_{i,j}^{t}$ is the water vapor at date t (master date) The initial water vapor map (t=0) is provided by a numerical weather model, in our case by the ERA-Interim model, and the subsequent dates (t=1,…,N) are calculated based on the previous water vapor solution. The following linear regression model with four parameters is proposed to introduce the information about the low frequency spatial distribution of water vapor provided by the numerical weather model into the time series of InSAR maps of PWV:

$$\begin{aligned}\mathrm{PWV}_{i,j}^{t+1} = \mathrm{PWV}_{i,j}^{t} &+ \alpha_1 \Pi \delta R_{i,j} \cos(\theta_{i,j}) \\ &+ \alpha_2 \varphi_{i,j} + \alpha_3\, \Lambda_{i,j} \cos(\varphi_{i,j}) + \alpha_4\end{aligned} \tag{9}$$

where $\alpha_{1,\ldots,4}$ are the parameters to be estimated, (φ, Λ) are the geodesic coordinates of pixel (i,j), and the $\cos(\varphi_{i,j})$ term is used to obtain isometrics coordinates. The four parameters are estimated using known PWV values, at least four locations that can be continuous operating GNSS stations [2], [3], [25]. It is worth noting that the model does not require a background water vapor map for every interferogram, but rather a small set of, at least, four continuous operating GNSS stations, necessary to correct the missing vertical reference and possible low-spatial-frequency error patterns introduced by the unwrapping algorithm. A linear model was chosen, because it shows less distortion in large zones without GNSS stations coverage, compared with higher degree models. Clearly, the best configuration is achieved with GNSS stations located close to each one of the corners of the interferometric area (something that is not always possible), capturing the spatial tendency over the entire area. To solve (9), a minimum of four GNSS PWV observations at SAR slave date are needed to find a solution and, in this case, the solution is unique. In the sense of the least-square approach, we can rewrite (9) as

$$Y_{n\times 1} = A_{n\times m} x_{m\times 1} + \epsilon_{n\times 1}. \tag{10}$$

where Y is the analysis state vector, A is the coefficients matrix, x is a vector of unknown parameters, ϵ is a vector of unobserved disturbances, and the subscripts refer to the matrix dimension, i.e., n refers to the observation number (in rows) and m refers to the variable number (in columns). For better illustration of this concept, (11), as shown at the

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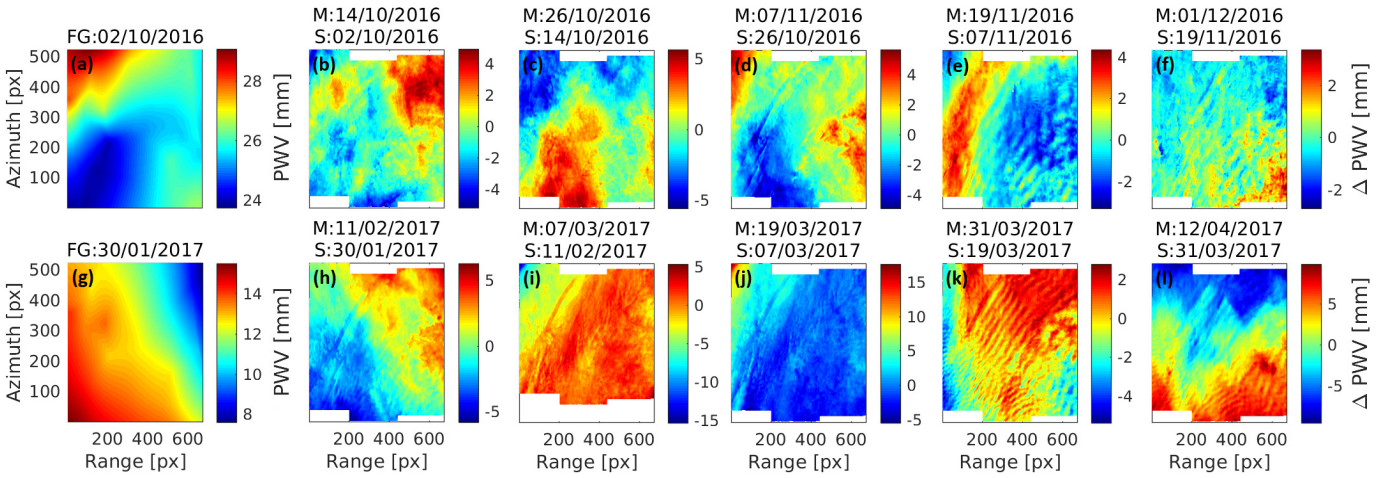


Fig. 2. (a) PWV first guest obtain from the ERA-Interim data on October 2, 2016, temporal interpolated to obtain the same SAR images acquisition time. (b)–(f) Uncalibrated ΔPWV maps estimated from the interferometric phase after applied the unwrapped algorithm, between October 2 and December 1, 2016. (g) PWV first guest obtain from the ERA-Interim data on January 30, 2017. (h)–(l) Uncalibrated ΔPWV maps estimated from the interferometric phase, between January 30 and April 12, 2017. Note that the map in (i) does not have the same size as the other maps, which conditions the size of the following maps in the iterative process (see Fig. 3).

bottom of the this page. defines the information of each vector and matrix in (10). In (11), the superscript "gnss" refers to the GNSS PWV values and "insar" refers to the InSAR PWV values at the GNSS stations location (φ, Λ). Due to the different acquisition geometries, between SAR (tube) and GNSS (cone), the InSAR PWV values at the GNSS location are calculated using the methodology proposed by Hanssen [15]. The least-square solution minimizes the sum of squared residuals, and leads to a closed-form expression for the estimated value of the unknown parameter vector x. The solution of (10) is given by

$$\hat{x}_{m\times 1} = \left(A^T_{n\times m} C_{n\times n} A_{n\times m}\right)^{-1} A^T_{n\times m} C_{n\times n} Y_{n\times 1}. \tag{12}$$

The vector of residuals can be computed from the data and the vector of estimates $\hat{x}$ by means of

$$\epsilon_{n\times 1} = Y_{n\times 1} - A_{n\times m}\hat{x}_{m\times 1}. \tag{13}$$

C in (12) is the covariance matrix of the observations errors, which is a square, symmetric with variances along the diagonal. The off-diagonal terms are cross-covariances between each pair of observations. In this article, we consider that the GNSS stations are sufficiently far apart in space so that the cross-covariances between them is zero. Thus, the diagonal of C is filled by the square of the precision of each GNSS PWV (determined by the GAMIT/GLOBK package), as

$$C_{n\times n} = \operatorname{diag}\left(\sigma_1^2, \sigma_2^2, \sigma_3^2, \ldots, \sigma_n^2\right). \tag{14}$$

The overall assessment of the estimated $\mathrm{PWV}^{t+1}_{i,j}$ is calculated using the following formulation

$$\sigma_{\mathrm{PWV}^{t+1}_{i,j}} = \left[\sigma^2_{\mathrm{PWV}^{t}_{i,j}} + \sigma^2_{\mathrm{InSAR}} + \sigma^2_{e_{1\ldots n}}\right]^{1/2} \tag{15}$$

where $\sigma^2_{\mathrm{PWV}^{t}_{i,j}}$ is the initial state error estimation, the second term is the InSAR PWV intrinsic error, and $\sigma^2_{e_{1\ldots n}}$ is the overall quality of the fit given by $\sum(e_1^2/(n-2), e_1^2/(n-2), \ldots, e_n^2/(n-2))$ [26].

V. Results

In this section, we summarize the results obtained in this work by applying the proposed methodology to the two time series of interferograms generated from Sentinel-1 data. The ΔPWV maps estimated from the interferometric phase can be seen in Fig. 2. The top row shows the ΔPWV maps, with a 12-day temporal baseline, obtained from SAR images acquired between October 2 and December 1, 2016, with the initial state on October 2. The bottom row shows the ΔPWV maps, also with 12 days temporal baseline (except for the first image pair), obtained from SAR images acquired between January 30 and April 12, 2017, with the initial state on January 30. Both initial states (first-guest) can be seen in Fig. 2(a) and (g). The temporal sequence of PWV maps is presented in Fig. 3. The triangles show the GNSS stations' locations, where the stations identified by their name are considered fixed in the least-square system. The color within the triangles is in the same color scheme and represents the estimated GNSS PWV at each station location. In a first visual inspection, a good agreement

$$Y = \begin{bmatrix} \mathrm{PWV}^{t+1}_{\mathrm{gnss}}(\varphi,\Lambda)_1 - \mathrm{PWV}^{t}(\varphi,\Lambda)_1 \\ \mathrm{PWV}^{t+1}_{\mathrm{gnss}}(\varphi,\Lambda)_2 - \mathrm{PWV}^{t}(\varphi,\Lambda)_2 \\ \vdots \\ \mathrm{PWV}^{t+1}_{\mathrm{gnss}}(\varphi,\Lambda)_n - \mathrm{PWV}^{t}(\varphi,\Lambda)_n \end{bmatrix}, \quad A = \begin{bmatrix} \Delta\mathrm{PWV}_{\mathrm{insar}}(\varphi,\Lambda)_1 & \varphi_1 & \varphi_1\cos(\Lambda)_1 & 1 \\ \Delta\mathrm{PWV}_{\mathrm{insar}}(\varphi,\Lambda)_2 & \varphi_2 & \varphi_2\cos(\lambda)_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ \Delta\mathrm{PWV}_{\mathrm{insar}}(\varphi,\Lambda)_n & \varphi_n & \varphi_n\cos(\lambda)_n & 1 \end{bmatrix}, \quad x = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}. \tag{11}$$

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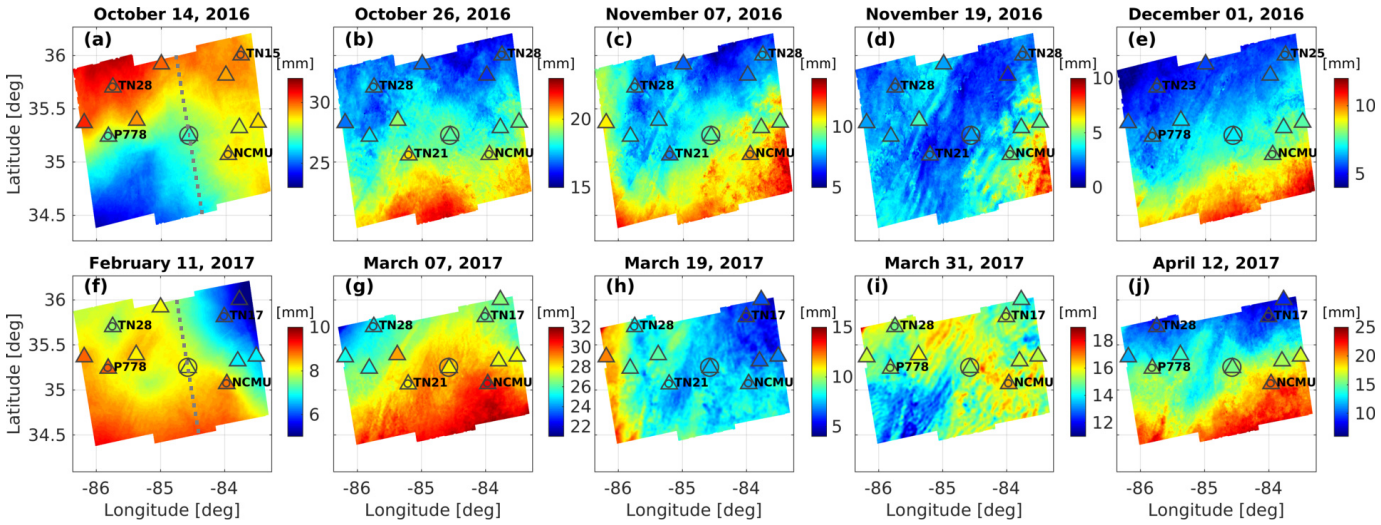


Fig. 3. (Top) PWV maps derived from a time series of interferograms with initial state in October 2, 2016, and a temporal baseline of 12 days. (Bottom) PWV maps with initial state in January 30, 2017, and a temporal baseline of 12 days except for the first image pair. Triangles: GNSS location, the color within are in the same color scheme. The stations identify by their name are considered fixed. The dashed line in (a),(f) and the TN29 GNSS station identify by the triangle within a circle reveals the south-north location of the profile (see Fig. 4).

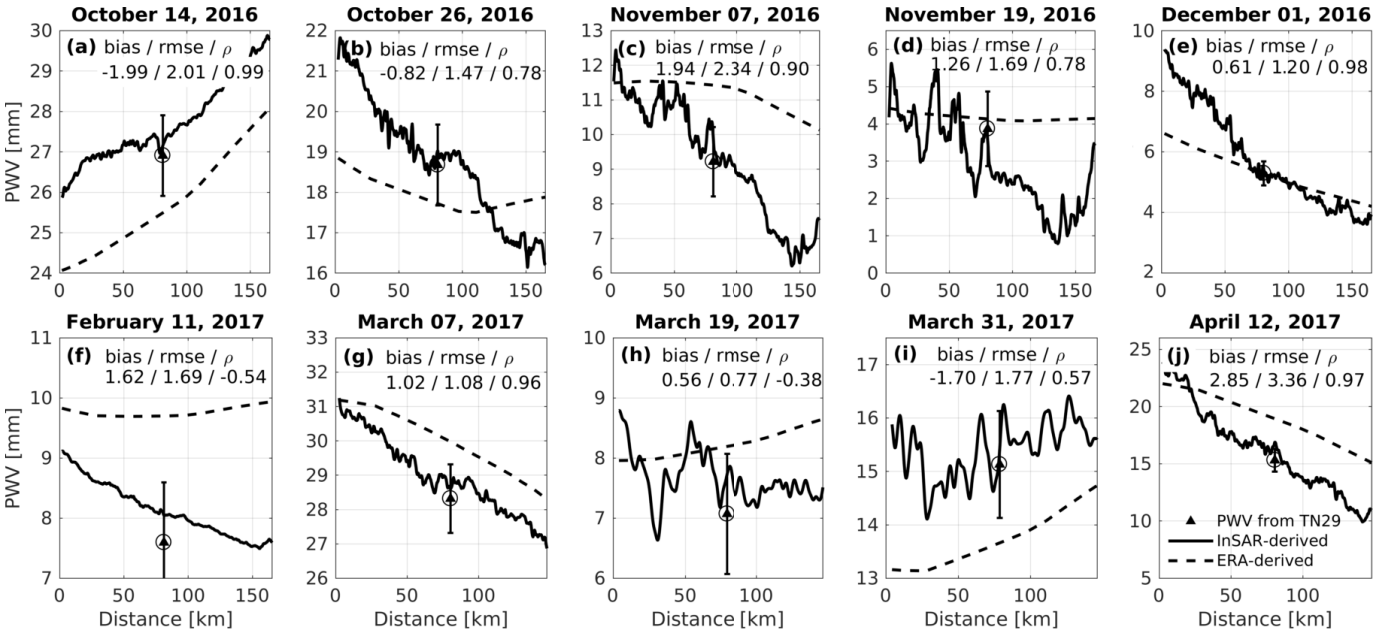


Fig. 4. (Top) North–northwest profile of PWV derived from InSAR thick and ERA-Interim dashed for the first time series. This profile passes over the station TN29 (triangle within a circle, see Figs. 1 and 3). Bottom: same information for the second time series.

between GNSS PWV values and the corresponding values in the InSAR PWV maps (and pixels surrounding) is observed, with a mean RMSE of 0.65±0.16 mm. About 75% of the PWV maps show gradients in the north–south direction, which can be mainly explained by the transition between the subtropical zone in the south and the temperate zone on the north. Fig. 4 shows a north–northwest profile of PWV derived from InSAR and ERA-Interim, crossing by the GNSS station TN29 (triangle within a circle, see Figs. 1 and 3). As expected, the PWV profile derived from InSAR data exhibits much more spatial detail when compared with the ERA-Interim profile. In fact, the ERA-Interim PWV product is obtained using a bilinear interpolation algorithm, applied to the ERA-Interim native resolution of 0.75°×0.75° [24]. Thus, we only evaluate the spatial tendency using the Pearson correlation coefficient (ρ). Strong correlation coefficients were obtained, varying between 0.58 and 0.99, except on March 31, 2017. A comparison with a high-resolution NWP models helps us to conclude that the PWV oscillation signature, which can be seen in Figs. 3(i) and 4(i), are attributed to atmospheric gravity waves (AGW), a very localize phenomena associated with complex topography only characterized at mesoscales and microscales. This phenomenon, not detectable at the ERA-Interim scale, can explain the lack of correlation on this day. Table I summarizes the bias, RMSE, and correlation coefficient for GNSS versus ERA-Interim and GNSS versus InSAR (in this evaluation only, the nonfixed stations were used). In an overall analysis, the ERA-Interim shows a bias of 0.52±1.76 mm,

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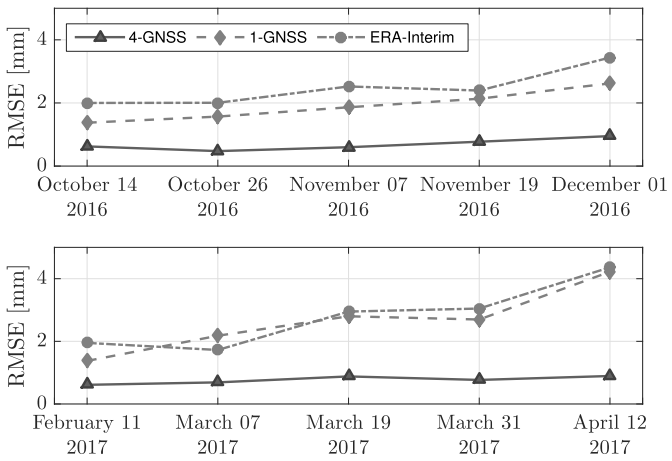


Fig. 5. RMSE between the results obtained by the proposed methodology (4-GNSS) and two different approaches. 1-GNSS: only 1 GNSS station (TN29) to calibrate each interferogram. ERA-Interim: fields of PWV simulated by ERA-Interim at each date to obtain the PWV maps.

TABLE I

STATISTICAL ANALYSIS. BIAS, RMSE, AND CORRELATION COEFFICIENT (ρ) FOR GNSS VERSUS ERA-INTERIM AND GNSS VERSUS INSAR. THE BOTTOM TWO ROWS SHOW THE MEAN AND STANDARD-DEVIATION OF ALL COLUMNS

Date	ERA-Interim			InSAR		
	bias	rmse	ρ	bias	rmse	ρ
October 14, 2016	-2.81	2.94	0.71	-0.17	0.56	0.89
October 26, 2016	-0.32	2.08	-0.22	0.13	0.50	0.87
November 07, 2016	2.02	2.43	-0.03	0.22	0.49	0.93
November 19, 2016	0.78	1.35	-0.35	0.03	0.79	0.65
December 01, 2016	0.53	2.15	0.65	0.15	0.82	0.91
February 11, 2017	1.16	1.38	0.36	0.10	0.45	0.95
March 07, 2017	0.75	1.56	0.58	-0.04	0.63	0.89
March 19, 2017	1.05	1.29	0.93	-0.30	0.58	0.97
March 31, 2017	-1.48	1.72	0.35	-0.06	0.80	0.38
April 12, 2017	3.56	3.87	0.85	-0.23	0.91	0.96
mean	0.52	2.07	0.38	-0.02	0.65	0.84
standard-deviation	1.76	0.82	0.44	0.17	0.16	0.18

an RMSE of 2.07±0.82 mm, and a correlation coefficient of 0.57±0.44, overestimating the PWV in 60% of the cases and underestimating in 40%. The first time series shows a lack of correlation between October 26 and November 19, mainly due to the smoothness of the PWV filled simulated by the ERA-Interim. Nonetheless, the spatial gradient was well captured as can be seen in Fig. 4(b)–(d). In the InSAR case, the bias was reduced by 96% (−0.02 mm), the RMSE by 69% (0.65 mm) and the correlation coefficient increased by 55% (0.84), comparatively to the ERA-Interim statistical analysis. The error of the initial state $\sigma^2_{\mathrm{PWV}^t_{i,j}}$, for each iteration, was calculated against the nonfixed GNSS stations data, and we also assumed that the InSAR PWV intrinsic error is about 1 mm, a reasonable value to include interferometric processing errors and errors due to the approximation in Π [see (7)]. The overall assessment, calculated using (15), shows values between 1.1 and 1.5 mm for the first time series and between 1.4 and 1.7 mm for the second one. We have also compared the results of the proposed methodology with two particular cases: 1) when only one GNSS station (TN29) is available in the area and 2) when all interferograms are using the ERA-Interim data as background water vapor map. In the first case, $\alpha_1 = 1$, $\alpha_{2,3} = 0$ and $\alpha_4 = \mathrm{PWV}^{\mathrm{gnss}}_{i,j} - \Delta\mathrm{PWV}^{\mathrm{insar}}_{i,j}$, and in the second case, $\alpha_1 = 1$, $\alpha_{2,3} = 0$ and $\alpha_4 = \mathrm{PWV}^{era}_{i,j} - \Delta\mathrm{PWV}^{\mathrm{insar}}_{i,j}$. In both cases, the initial state is constrained with ERA-Interim and GNSS data. Fig. 5 shows the RMSE, obtained from comparing the independent GNSS PWV values with the corresponding PWV values in each date. As expected, the RMSE tends to increase with time using only one GNSS station or ERA-Interim fields. However, both have similar behavior over time. In areas with few (or a lack of) GNSS stations, it is possible to use one of the last two mentioned approaches for short time series.

VI. CONCLUSION

In this article, we present a methodology to generate times series of high-resolution spatial variations of PWV using InSAR data. The proposed algorithm shows some advantages compared to other methods in the literature. To start with, the reference ERA-Interim water vapor map can be chosen at a time also different from the master acquisition time. As a consequence, it is possible to select the ERA-Interim map at a time with no turbulent propagation into the atmosphere. Furthermore, there is no need for an interferometric stack, and it is not necessary to constrain the model with assumptions about the time series statistics. Finally, computational resources are no longer a concern, and the initial state can be derived from ERA-Interim data at any location. It is worth noting that the initial state can be obtained from the NCEP-DOE Reanalysis 2 or from the new ERA5 reanalysis model that provides hourly estimates of a large number of atmospheric, land and oceanic climate variables at 30-km global grid (ERA5 will eventually replace the ERA-Interim reanalysis). The initial state can be defined on any SAR image date; however, this date should be selected carefully, in order to identify the cases when the NWP data does not describe the atmospheric water vapor spatial gradients. In our experience, the choice falls on the interferogram (master or slave date) that shows a stratified water vapor. In the overall assessment, we found values lower than 2 mm [see (15)] for both time series over the complex region chosen for this study.

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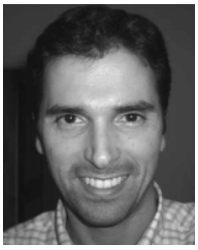
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